Certificate Price Test - NSF and Moonshots

TLDR: Purpose of this experiment is to test prices for NSF and Moonshots via two separate experiments. We have seen strong evidence to suggest a large opportunity to charge more for both certificates. We're aiming at a big swing here, we'd need a 15% impact to see a high confidence decision.

As of:	Oct 7, 2025
Risk Score	🔴
Status	☑ ~~Scoping~~ ☑ ~~Designing~~ ☑ ~~Building~~ ☑ ~~Live~~ ☐ Analyzing ☐ Concluded
RAPID	**Recommend**: Kaison, Mel, Benjamin, Annie **Agree**: Mandar, Robbie, Valen **Perform**: Certs Pod **Inform**: **Decide:** Taylor
Links	Jira | Looker Dashboard | Amplitude Dash | Tech Spec | Event Spec Conjoint Study | Price Test Stat Feasibility Analysis Price Sheets: $299 | $599 | $999 NSF Force links: control | variant 1 | variant 2 NSF: Looker Dashboard (Looker Dash NSF orders only) | Amplitude Dash Moonshots: Looker Dash (Looker Dash Moonshots orders only)| Amplitude Dash
Experiment Flag Name	Need two separate experiment flags here NSF: exp_mccerts-pricing-leadership-and-decision-making-under-pressure Moonshots:

	mccerts-pricing-breakthrough-innovation-strategy-from-bold-idea-to-business-impact
Key Dates	Statistical Feasibility Analysis: Mar 20, 2025 Build Start: Oct 6, 2025 Moonshots Go/No-Go: Oct 20, 2025 QA Start (target): Launch (target): Nov 3, 2025 ⇒ drop dead, earlier is better Experiment Launch (actual): Oct 28, 2025 , 9 am End: Nov 19, 2025 Promos Start: Nov 20, 2025

Pending Decisions / Action Items

- Which geos do we expect to push between 11/3 and 11/20?
  - Currently for moonshots, just targeting US and other T1 countries
  - NSF only for US.
- Do we think we can get 45k traffic between 11/3 and 11/20 for Moonshots AND NSF?
  - 10/20 confirm we're on track for Moonshots traffic
- Reactive Comms
- Event Spec
- DECISION Oct 8, 2025 : the earliest possible date for a certs promo will be 11/20.

Guardrails

1. **Only test prices higher than $299**: we want to maintain a delta between certs and core, and evidence suggests highest upside to increasing prices.
2. **Do right by the customer:** Customers might be understandably frustrated if they purchase a higher price point and then see afterwards a lower price.
3. **Capture the bookend prices**: make sure that we capture a wide enough range of prices to determine the optimal price within that range.

What Prices do we test?

Scenarios	Pros	Cons
Draw the Best Curve: $299-$599-$999 RECOMMENDATION	This allows us to draw a curve that covers a wider range from $299 to $999. - There's a risk here that the confidence intervals of each tested price point might	It is possible (slim) that the $999 price point would drive significant negative impact and we would either have to pause that variant or accept the revenue loss here.

HERE	overlap, which would mean less confidence in determining ideal price within the range of highest and lowest.	
Conservative: $299-$599-$799	We have higher confidence in this setup as data suggests we can go up to $799 before seeing slowed rev growth	We miss out on a higher price point for Certs beyond what the conjoint data suggests before slowed rev gains.

Risks

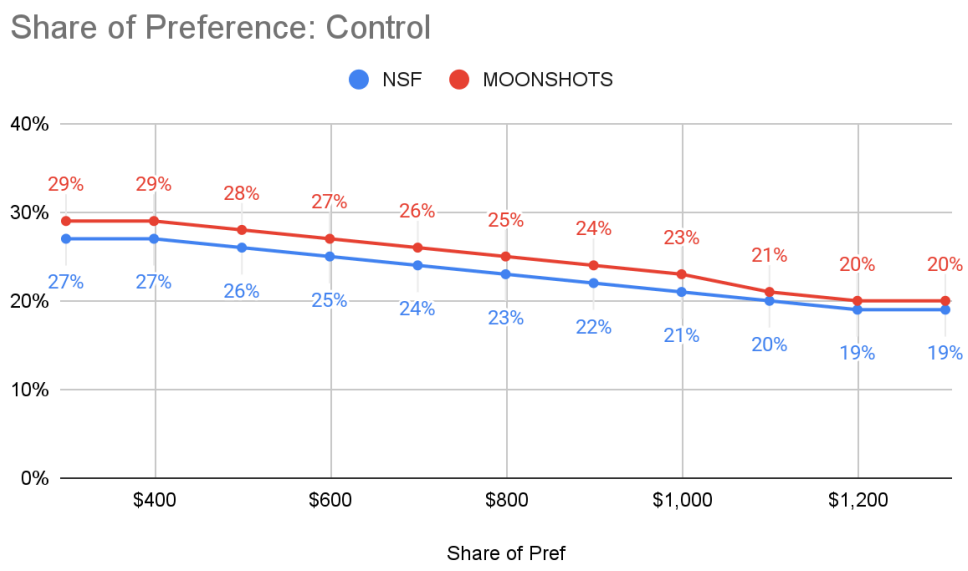
Risk	How We Address In this Experiment
PR announcement contradictions	Wait at least 2 weeks after announcement. 🟡 11/3 launch date is 1.5 weeks out from 10/23 launch for Moonshots. Confirm with Marketing here, Tawnya Bear confirms Oct 8, 2025
Ad Layer Thrash (observed at each price test on core)	Ensure revenue gets sent back to platform via gold standard event properties with new SKU. Rohit Thosar can help test values here once events implemented. Max conversions or value based? For now, max conversions.
Underpowered Test with <20% sales/visitor	We will design the test with 20% MDE, 80% Power, 10% alpha. We will need 45k traffic on each page to hit sample size. We will only make a call on confident 'home runs'
Users become aware of price points being tested: - During consideration they are bucketed into multiple variants (high price point, long consideration cycle) or it leaks on socials. - After purchase	**Support Appeasement** ⇒ If users reach out, we can force link them to a lower price point, or issue a partial refund if after purchase. *Core learnings: when we test %s for core promo, we defaulted users to best discount if they reached out. When we tested core prices, we upgraded users to higher tier for those who reached out.* **Social Listening**, like Dash Social (organic) or Sprinkler (paid) to detect user conversations / comments on core social channels. Rohit Thosar can help set this up for monitoring. Robbie Giles and Rohit Thosar to discuss who. **We will not:** - Match market by state, region, zip. Does not look like metro/state/IP level detection is consistent per user (Sheel).
Experiment Results come in right before Holiday 11/20	During experiment duration, have frequent checks on data so that we are prepared to make a call on 11/19 to flip 100% of traffic to winning bucket. Early Stopping Rules Ensure that promo works with this price test (ie if we flip switch to variant 1, make sure promo

	flows work).

Experiment Justification

Problem / Opportunity	Pricing
Other Quotes or Evidence	**Key Learnings from an [August '25 Conjoint Study]** 1. We see significant double digit rev gains from $299 to $799 2. See the biggest dip on preference after $899 (suggesting we should not go over $900) 3. See the rev growth go into single digits after $799 (suggesting we should not go over $800)

REVENUE X 100	NSF	MOONSHOTS	
$299	$11,765	$12,283	
$399	$15,389	$15,993	31%
$499	$18,899	$19,621	23%
$599	$22,142	$23,001	17%
$699	$25,007	$25,991	13%
$799	$27,804	$28,868	11%
$899	$30,127	$31,368	8%
$999	$31,225	$32,515	4%
$1,099	$31,339	$33,238	0%
$1,199	$33,357	$35,212	6%
$1,299	$35,148	$37,004	5%



Hypothesis	If we run a price test on each certificate after launch, we'll optimize sales per visitor and we'll capture significant revenue upside because demand seems inelastic to price.
Impact Estimate / Size of Opp	20% increase in sales/visitor can drive an extra $500k in Q4
Primary Metric	Sales/Visitor
Secondary Metrics	CVR
Counter Metrics	Refund Rates
Surface Area(s)	NSF Checkout Flow Moonshots Checkout Flow
Run Time Estimate	45k traffic needed per product page with 20% MDE, 10% Alpha. 14k traffic per variant <u>As of 10/7:</u> **NSF**: trailing 7-day traffic 20k ⇒ 2.5 weeks **Moonshots**: trailing 7-day traffic 8.3k ⇒ 4-5 weeks

Input known values

		9/4 - 9/24	9/4 - 9/24	
Baseline CVR	1.38%	Aggregated total (	NSF CVR	
Certificate Value	$299	AOV		
Baseline Sales Per Visitor	$4.1			249
Baseline Revenue	$397,462			3.4362
SPV STD	28.89			
Alpha	0.1			
Power	0.8			
Z-crit (alpha) (2 tail)	1.6449			
Z-crit (beta)	0.8416			
Weekly Traffic	24,000	weekly traffic from first 4 weeks		
Daily Traffic	3,429			
Total Variants (including Control)	3	<- adjust variations		

Sample Size Calculator for Sales per Visitor

Priced at $299						Adjust MDE value here to see custom runtimes.
MDE (Sales per Visitor)	5%	10%	15%	20%	30%	40%
Hypothetical Sales / Visitor	$4.33	$4.54	$4.75	$4.95	$5.36	$5.78
Hypothetical Revenue	$103,980	$108,932	$113,883	$118,835	$128,737	$138,640
Effect Size	0.007	0.014	0.021	0.029	0.043	0.057
Sample Size Needed Per Variant	242,467	60,617	26,941	15,154	6,735	3,789
Total Sample Size Needed	727,401	181,850	80,822	45,463	20,206	11,366
Days Needed	212.2	53.0	23.6	13.3	5.9	3.3
Weeks Needed	30.3	7.6	3.4	1.9	0.8	0.5

Note for Early Stopping: **Early Stopping Rules**

- Run for at least a week.
- At 25% of total sample size (can be 20% if traffic is really slow): p-value <0.01.
    - At least 3,500 visitors per variant, 20% MDE or higher, 99% significance

Input known values

		9/4 - 10/30		
Baseline CVR	1.46%	Aggregated total CVR		
Certificate Value	$299	AOV		
Baseline Sales Per Visitor	$4.4		249	9/4 - 9/24
Baseline Revenue	$397,462		3.6354	NSF CVR
SPV STD	28.89			
Alpha	0.1			
Power	0.8			
Z-crit (alpha) (2 tail)	1.6449			
Z-crit (beta)	0.8416			
Weekly Traffic	6,000	expected weekly traffic (9/4 - 10/30)		
Daily Traffic	857			
Total Variants (Including Control)	3	<- adjust variations		

Sample Size Calculator for Sales per Visitor

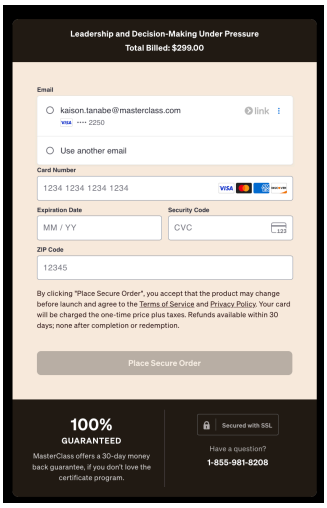
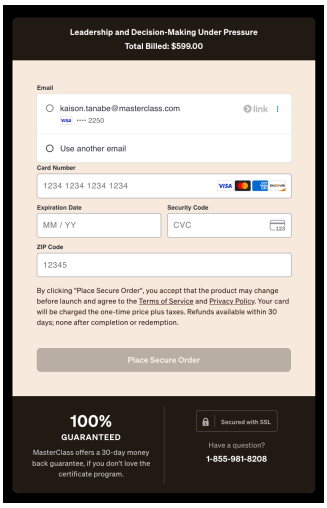
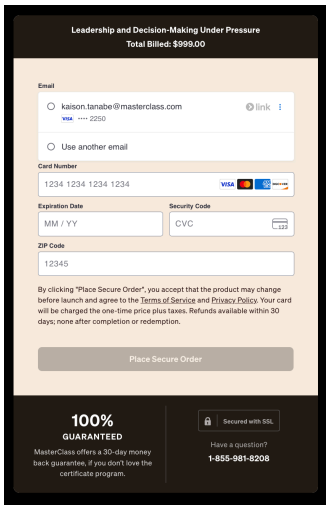
Priced at $299						Adjust MDE value here to see custom runtimes.
MDE (Sales per Visitor)	5%	10%	15%	20%	30%	40%
Hypothetical Sales / Visitor	$4.58	$4.80	$5.02	$5.24	$5.68	$6.11
Hypothetical Revenue	$27,502	$28,812	$30,121	$31,431	$34,050	$36,669
Effect Size	0.008	0.015	0.023	0.030	0.045	0.060
Sample Size Needed Per Variant	216,623	54,156	24,069	13,539	6,017	3,385
Total Sample Size Needed	649,870	162,467	72,208	40,617	18,052	10,154
Days Needed	758.2	189.5	84.2	47.4	21.1	11.8
Weeks Needed	108.3	27.1	12.0	6.8	3.0	1.7

Traffic above based on PC traffic (given EV fires on PC)

- Revised PC Traffic expecting 3.5k traffic for NSF alone

Experiment Design

	Control	Variant 1	Variant 2
Traffic %	33.33%	33.33%	33.33%
Description	$299	$599	$999

Design (screenshots)			

Engineering (requirements, acceptance criteria, other)

- Actually charge the users the higher price point (THIS IS NOT A PAINTED DOOR)
- Logged Out Users, Logged in Core Members who see the 15% discount (includes storefront experiment users)
- Geos:
  - All Geos (testing becomes longer as we have 3 sheets)
  - ~~Open question with Jhonathan Rath and Jon Pease which geos we expect traffic to come from, currently 90% of NSF purchasers are US (amp)~~
- Desktop/Mobile
- Exclude anyone who comes in via a partner vanity URL (ie Google Moonshots URLs)
- Currently masterclass at work emails can't purchase certs.
- New SKUs for each variant
  - All Gold Standard Events reflect the new SKU properties (ie price, SKU, etc)
- Promotion dynamic starting 11/20 must work with control or either variants
- Separate experiments for Moonshots and NSF
- Ability to force link users into a bucket (support appeasement)

Events and Analytics Requirements

Benjamin Lee : EV and adjustments to Order Completed for the different SKUs are the main things needed. Should be good to track in the experiments dashboard from there

Event Spec (Gold Standard Here)	Link to Confluence Spec

Analytics Requirements + Questions	P0 / P1 / P2 *Ie I want to do a funnel analysis for all pages, or how many people click on a carousel, and slice data by instructor, etc.* **P0 - Experiment Viewed** Ensure all properties correlate to new SKU for each new price **P0 - Order Completed** Ensure all properties correlate to new SKU for each new price

NSF Results (data and analysis with one-liner)

Date: 10/28 (9:30AM PST) - 11/3 (MDE %)
Results PDF: Purchases NSF Price test [Dashboard]

TL;DR

🔑 Sales per visitor dropped significantly (-31% at $599 and -45% at $999), with a 10-point higher drop-off after viewing price in variants versus control. End the test early and keep control to maximize holiday revenue.

What was tested

- Testing price variants for NSF with Control at $299 and Variants of $599 and $999 as determined by our [Conjoint Study] & [Price Test Stat Feasibility Analysis]

Results takeaways

- -31% and -45% (stat sig) on Sales / Visitor respectively for $599 and $999
- Saw a 10ppt increase in dropoff after viewing the price in Variants compared to Control.
- Purchase trends seem directionally flat / negative for both Variants and doesn't seem likely that the sales/visitor lift will surpass Control

Recommendations

- Key decision by execs to end the test early given early negative signals. [(slack link)]
- End early, keep control to maximize revenue for upcoming Holiday season

Supporting Data

1. **Results Summary** (11/3):

Significant impact to Sales per Visitor in Variant 2 (-45%) with Variant 1 close behind (-31%)
CVR is down as expected given higher price points.

At this point we are at 6% of required sample size to call the experiment.

CVR								
variation_name	visitors	Purchasers	CVR Mean	CVR Lift (%)	CVR Significance	CVR CI Lower	CVR CI Upper	CVR P value
control	821	42	5.12%	0.00%	100.00%	NaN	NaN	NaN
variant_1	817	15	1.84%	-64.11%	99.99%	-86.68%	-41.54%	0
variant_2	860	7	0.81%	-84.09%	100.00%	-104.82%	-63.35%	0

Sales / Visitor									
variation_name	visitors	Sales/Visitor Revenue	Sales/Visitor Mean	Sales/Visitor Std	Sales/Visitor Lift (%)	Sales/Visitor Significance	Sales/Visitor CI Lower	Sales/Visitor CI Upper	Sales/Visitor P value
control	821	$12,095	14.7	63.8	0.00%	100.00%	NaN	NaN	NaN
variant_1	817	$8,259	10.1	74.2	-31.38%	91.16%	-61.13%	-1.63%	0.177
variant_2	860	$6,907	8.0	88.9	-45.48%	96.25%	-78.20%	-12.76%	0.075

2yr LTV / Visitor									
variation_name	visitors	2yr LTV/Visitor Revenue	2yr LTV/Visitor Mean	2yr LTV/Visitor Std	2yr LTV/Visitor Lift (%)	2yr LTV/Visitor Significance	2yr LTV/Visitor CI Lower	2yr LTV/Visitor CI Upper	2yr LTV/Visitor P value
control	821	$9,676	11.8	51.0	0.00%	100.00%	NaN	NaN	NaN
variant_1	817	$6,608	8.1	59.4	-31.38%	91.16%	-61.13%	-1.63%	0.177
variant_2	860	$5,526	6.4	71.1	-45.48%	96.25%	-78.20%	-12.76%	0.075

Certificates - Summary Results

Experiment Name	Variation Name	Visitors	Purchasers	CVR	Refunded Purchases	Purchases Refund Rate	Sales per Visitor	2yr Ltv per Visitor	5yr Ltv per Visitor	AOV	Sales	CVR % Change	CVR Significance	RR % Change	RR_Significa	Sales Per Visitor %Change	Sales Per Visitor Significance	2yr LTV/Visitor %Change	2yrLTV Per Visitor Significance	5yr LTV/Visi %Change
mccerts-...	control	919	50	5.44%	1	1.96%	$16.62	$13.30	$13.30	$299.57	$15,278	0.00%	50.00%	0.00%	50.00%	0.00%	50.00%	0.00%	50.00%	0.00%
mccerts-...	variation...	923	20	2.17%	2	10.00%	$10.31	$8.25	$8.25	$475.63	$9,513	-60.17%	99.99%	410.00%	87.32%	-38.01%	97.10%	-38.01%	97.10%	-38.01%
mccerts-...	variation...	948	8	0.84%	1	11.11%	$7.33	$5.86	$5.86	$771.74	$6,946	-84.49%	100.00%	466.67%	80.30%	-55.93%	99.56%	-55.93%	99.56%	-55.93%

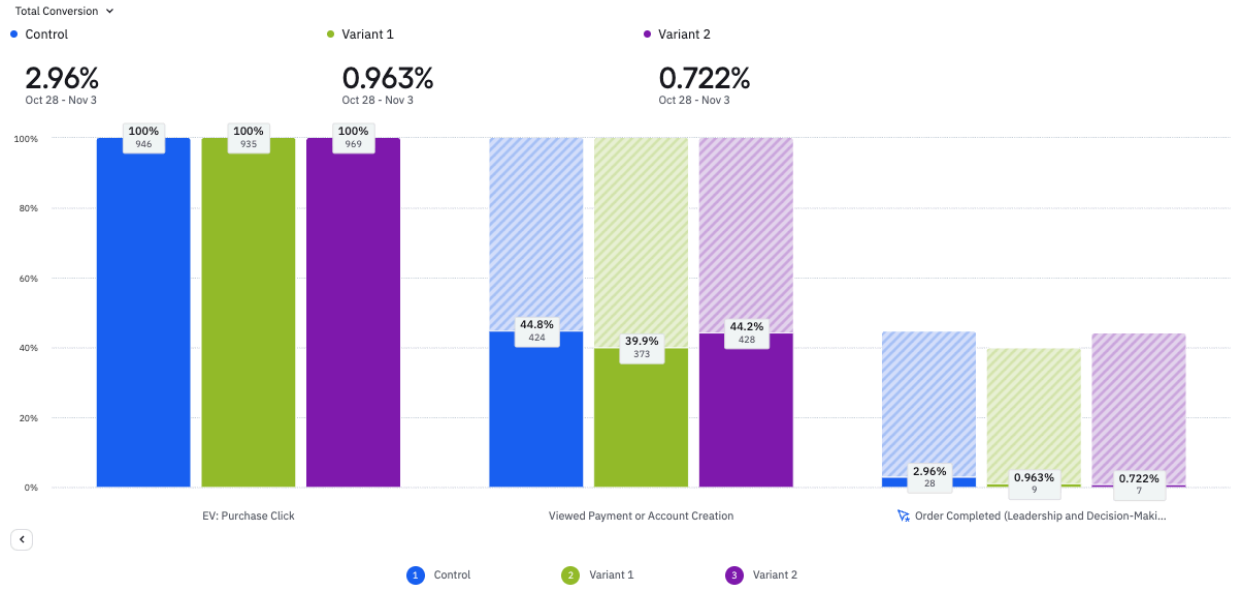
New vs Existing Users

- Similar to overall results, we see diminishing results as price increases

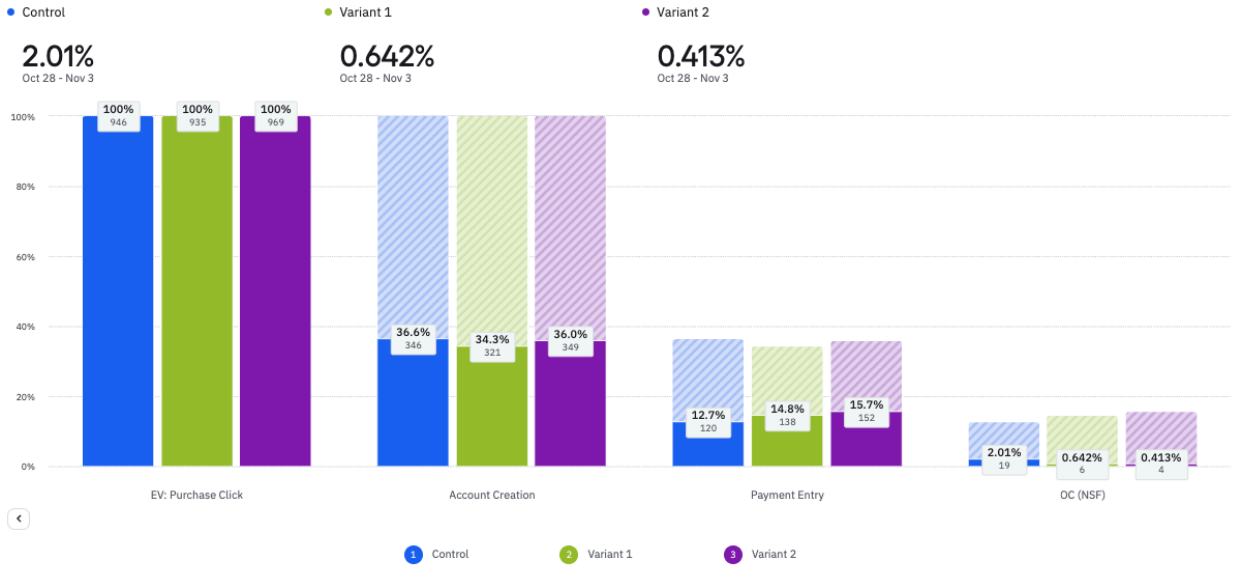
	new users			existing users		
variation_name	control	variant_1	variant_2	control	variant_1	variant_2
visitors	723	727	781	135	119	104
Purchasers	47	9	6	16	6	1
CVR Mean	6.50%	1.24%	0.77%	11.85%	5.04%	0.96%
CVR Lift (%)	0.00%	-80.96%	-88.18%	0.00%	-57.46%	-91.89%
CVR Significance	100.00%	100.00%	100.00%	100.00%	97.29%	99.94%
Sales/Visitor Revenue	$13,881	$4,972	$5,828	$4,291	$3,185	$1,079
Sales/Visitor Mean	19.20	6.84	7.46	31.78	26.76	10.38
Sales/Visitor Lift (%)	0.00%	-64.38%	-61.13%	0.00%	-15.79%	-67.36%
Sales/Visitor Significance	100.00%	99.97%	99.80%	100.00%	64.90%	95.17%
2yr LTV/Visitor Revenue	$11,105	$3,978	$4,663	$3,433	$2,548	$863
2yr LTV/Visitor Mean	15.36	5.47	5.97	25.43	21.41	8.30
2yr LTV/Visitor Lift (%)	0.00%	-64.38%	-61.13%	0.00%	-64.38%	-61.13%
2yr LTV/Visitor Significance	100.00%	99.97%	99.80%	100.00%	99.97%	99.80%

2. **Purchase Funnel (link1) (link2) :**

- Link1: Logged in MC Users skip the Account Creation page, so combining them inline to capture all users.
- Link2: has those steps split out
- Looking only at NSF purchases

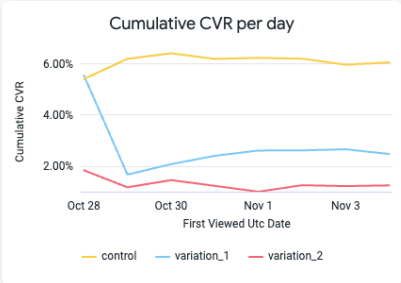


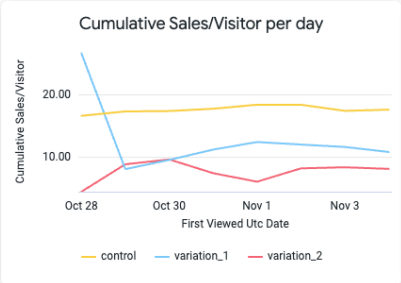
Looking at the segmented funnel, we see Control with 85% dropoff after viewing the Price, while Variant 1 & 2 have upwards of 95% dropoff after viewing the Price. ~5% increase in dropoff for Variants compared to Control

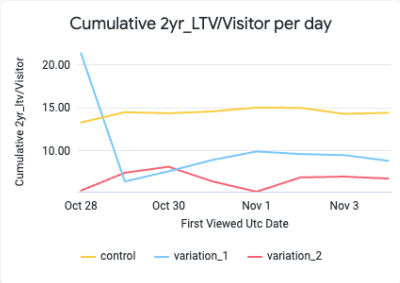


3. Purchase Trend Segments:

No clear upward trend for Variants 1 & 2 after 7 days







Moonshots Results (data and analysis with one-liner)

Date: 10/31 - 11/5 (MDE %)
Results PDF: [Dashboard]

TL;DR

🔑 Sales per visitor declined sharply (-55% at $599 and -38% at $999), with an 8-point higher drop-off after viewing price in variants versus control, and flat-to-negative purchase trends suggest sales won't surpass control. End early, keep control to maximize revenue for upcoming Holiday season

What was tested

- Testing price variants for Moonshots with Control at $299 and Variants of $599 and $999 as determined by our Conjoint Study & Price Test Stat Feasibility Analysis

Results takeaways

- -55% (stat sig) and -38% on Sales / Visitor respectively for $599 and $999
- Saw a 8ppt increase in dropoff after viewing the price in Variants compared to Control.
- Purchase trends seem directionally flat / negative for both Variants and doesn't seem likely that the sales/visitor lift will surpass Control

Recommendations

- Key decision by execs to end the test early given early negative signals. (slack)
- End early, keep control to maximize revenue for upcoming Holiday season

Supporting Data

1. **Results Summary**

Similar to NSF, we see stat sig negative impact to Sales / Visitor (> -40%) for both Variants

Certificates - Summary Results

Experiment Name	Variation Name	Visitors	Purchasers	CVR	Refunded Purchases	Purchases Refund Rate	Sales per Visitor	2yr Ltv per Visitor	5yr Ltv per Visitor	AOV	Sales	CVR % Change	CVR Significance	RR % Change	RR_Significa	Sales Per Visitor %Change	Sales Per Visitor Significance	2yr LTV/Visitor %Change	2yrLTV Per Visitor Significance
mccerts-...	control	460	19	4.13%	0	0.00%	$13.48	$10.79	$10.79	$310.07	$6,201	0.00%	50.00%	∅	∅	0.00%	50.00%	0.00%	50.00%
mccerts-...	variation...	445	5	1.12%	0	0.00%	$6.02	$4.81	$4.81	$535.61	$2,678	-72.80%	99.78%	∅	∅	-55.36%	96.00%	-55.36%	96.00%
mccerts-...	variation...	390	5	1.28%	1	20.00%	$8.33	$6.67	$6.67	$650.09	$3,250	-68.96%	99.55%	∅	86.25%	-38.18%	84.31%	-38.18%	84.31%

New vs Existing Users

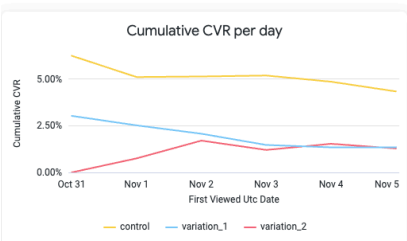
- Results to early to draw conclusions, existing users seem more likely to purchase higher price points

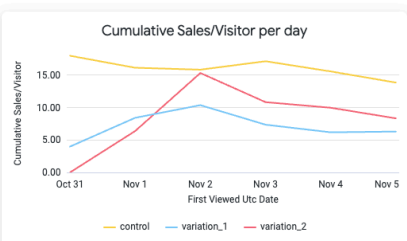
	new users			existing users		
variation_name	control	variant_1	variant_2	control	variant_1	variant_2
visitors	401	384	333	58	56	53
Purchasers	20	2	1	6	1	3
CVR Mean	4.99%	0.52%	0.30%	10.34%	1.79%	5.66%
CVR Lift (%)	0.00%	-89.56%	-93.98%	0.00%	-82.74%	-45.28%
CVR Significance	100.00%	99.99%	99.99%	100.00%	97.15%	81.68%
Sales/Visitor Revenue	$5,934	$1,126	$999	$1,648	$599	$2,697
Sales/Visitor Mean	14.80	2.93	3.00	28.41	10.70	50.89
Sales/Visitor Lift (%)	0.00%	-80.19%	-79.73%	0.00%	-62.35%	79.13%
Sales/Visitor Significance	100.00%	99.89%	99.62%	100.00%	87.34%	76.48%
2yr LTV/Visitor Revenue	$4,747	$901	$799	$1,318	$479	$2,158
2yr LTV/Visitor Mean	11.84	2.35	2.40	22.73	8.56	40.71
2yr LTV/Visitor Lift (%)	0.00%	-80.19%	-79.73%	0.00%	-80.19%	-79.73%
2yr LTV/Visitor Significance	100.00%	99.89%	99.62%	100.00%	99.89%	99.62%

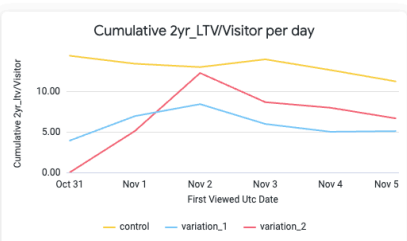
2. Purchase Trend Segments

Again seeing downward trends in the first 5 days of the experiment.

Daily and Segmented Metrics

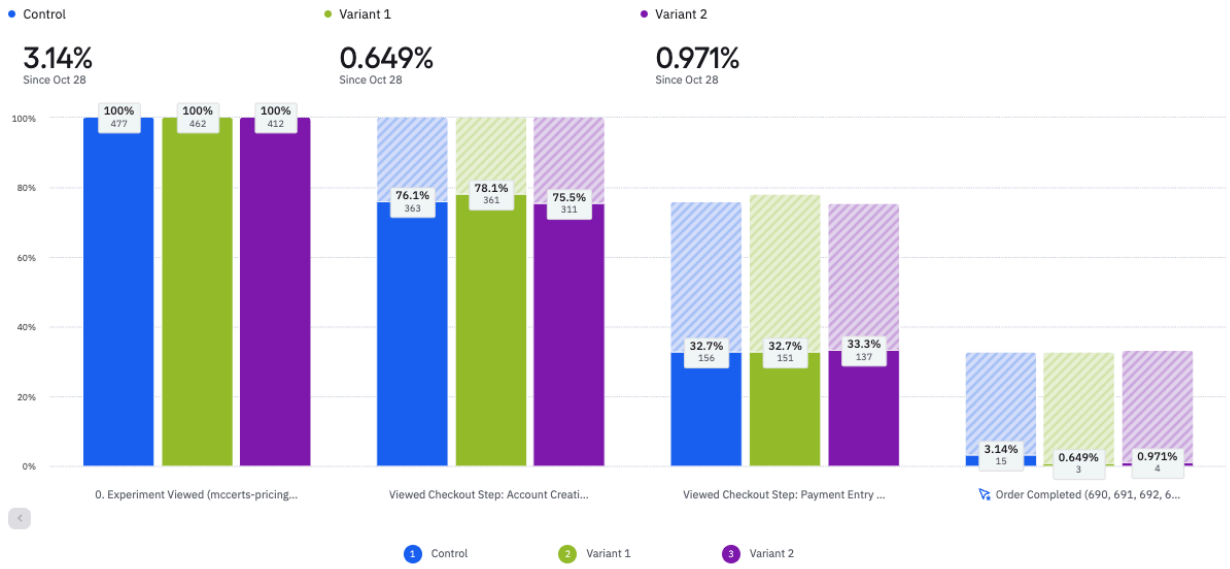






3. Purchase Funnel [(link)]

Where we saw 90% of users dropoff in Control after viewing the price, both Variants saw 98% users dropoff after viewing the price.



Notes:

- Mixed purchasers in Variations
    - Experiment Variations potentially not being tracked (EV not firing correctly)
    - Found $599 and $999 purchasers in Control, Standard purchasers in Variant 1
    - Logged in users get a different purchase flow experience vs logged out
- Sheet above assigns users right variation based on purchase
-

3:15:45 PM **Experiment Viewed**

mccerts-pricing-leadership-and-decision-making-under-pressure: variant_2

EVENT SPECIFIC PROPERTIES

experimentName: "mccerts-pricing-leadership-and-decision-making-under-pressure"

logged_in: false

page_source: "Certificates"

timings: {"experiment_fired_at":13759.10009765625,"page_interactive_at":350.80000019073486,"experiment_delay_after_interactive":13408.300097465515,"page_fully_loaded_at":9726.10000038147,"network_plus_response_time":196.2000002861023,"time_to_first_byte":169,"first_contentful_paint_at":452}

variationName: "variant_2"

GENERAL PROPERTIES

USER CONTEXT

PAGE CONTEXT

INTEGRATIONS CONTEXT

3:16:14 PM Product Added

3:16:14 PM Identify

3:16:14 PM Identify

3:16:14 PM Page Loaded

Checkout Step: Account Creation

3:16:15 PM Core Web Vital

TTFB: 315.3

3:16:20 PM Cart Email Captured

3:16:20 PM Payment Info Step Viewed

3:16:20 PM Identify

3:16:20 PM Page Loaded

Checkout Step: Payment Entry

3:16:14 PM **Product Added**

EVENT SPECIFIC PROPERTIES

```
page_source: "Certificate Checkout"

certificate_id: "1"
content_ids: [701]
content_type: "product"
currency: "USD"
name: "Leadership and Decision-Making Under Pressure"
price: 999
product: "certificates"
product_id: "701"
product_tier: "tier1"
product_tier_level: 1
quantity: 1
revenue: 999
```

GENERAL PROPERTIES >

USER CONTEXT >

PAGE CONTEXT >

INTEGRATIONS CONTEXT >

Competitor Inspo

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Notes

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